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Job embeddedness and life satisfaction during pandemic: the impacts of work-from-home stressors and emotional exhaustion

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Abstract

Purpose – This study applies the job demands-resources model the conservation of resources theory to explain and examine the impacts of home-work conflict, perceived insufficient organizational support, and perceived social isolation, that is, work-from-home stressors, o

Design/methodology/approach – Data were collected from 418 Thai employees who worked from home during the nationwide lockdowns at two time points. The hypotheses were tested and analyzed by means of a confirmatory factor analysis, structural equation modeling, and a bootstrapping procedure.

Findings – The results indicate that the three work-from-home stressors significantly cause emotional exhaustion and that these stressors are negatively associated with job embeddedness and life satisfaction via the mediation of emotional exhaustion.

Research limitations/implications – To reduce error in parameter estimation due to self-report data, future research could use a more rigorous longitudinal design with a longer time lag and collect data from multiple sources.

Practical implications – Realizing how critical situations shape the workplace would help organizations understand the issues concerning a remote work approach and create more applicable interventions to improve employees' retention and wellbeing.

Originality/value – This study reinforces the application of COR in times of crisis and extends the traditional JD-R model beyond the normal work context.

Keywords Crisis, Work-from-home, Emotional exhaustion, Job embeddedness, Life satisfaction, Thailand

Paper type Research paper

Introduction

The recent rapid worldwide spread of a novel coronavirus infection (COVID-19) has led to a global health crisis and an economic downturn. Governments imposed total lockdowns, banning unnecessary travel, and requiring the closure of all non-essential activities. The business and industry shutdowns implemented and mandated to control the spread of the virus have generated a wide array of unique challenges for both employees and employers (Ahlstrom and Wang, 2021). The strict control measures have caused many inconvenient working conditions. Normal ways of working have encountered serious challenges, and a broad range of remote-technology operations have become the new normal (Grant and Russell, 2020). The pandemic has made work-from-home (WFH) arrangements – the new way of working for millions of employees around the globe (Chen, 2021). Many workplaces are forced to adopt this approach by urging their employees to stay at home, practicing social distancing to limit face-to-face interactions with others, and switching, quite suddenly, to a remote work approach without proper planning to defend themselves against the outbreak. Thailand is no exception.

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WFH, also known as online working remotely, is defined as periodic work at home (away from a physical office location), one or more days a week (Hill *et al.*, 2003). Before the pandemic, the WFH model was recognized as a win-win scenario for both employees and employers. It has been considered an alternative way of organizing work to provide flexibility to workers in terms of hours, balancing work and nonwork responsibilities, and saving on commuting time (Felstead and Jewson, 2000). For businesses, several advantages of offering WFH include attracting and retaining highly skilled employees, as well as improving job satisfaction, quality of work-life, and even work performance (Bailey and Kurland, 2002; Crosbie and Moore, 2004).

However, since the outbreak, this means of working is being implemented widely, in an attempt to reduce the spread of the virus and to maintain work operations and employment during a period in which workplaces are closed and social distancing performed (Vyas and Butakhieo, 2021). WFH is then no longer regarded as an exclusive option, but a critical approach that must be taken by many businesses and workers to minimize health risks. This unprecedented phenomenon becomes one of the most remarkable changes that shifts many workers into this new normal across occupations and forces workplaces to adopt this way of working, without providing their employees with the necessary skills for remote work (Afrianty *et al.*, 2021; Chen, 2021). Many employees need to change not only their physical behaviors but also their work mentalities. It is argued that while WFH offers certain benefits, it also has some significant drawbacks. Prior studies have shown that employees are faced with various stressful events and demands (i.e. abrupt changes to the online work approach, complex work arrangements, being on continuous call, family activities) while staying at home and remotely working (e.g. Belzunegui-Eraso and Erro-Garcés, 2020; Chen, 2021; Syrek *et al.*, 2022). They have encountered negative impacts of WFH, such as a lack of social interaction, a lack of support from their employers or teams, balancing the demands of family and personal life, and occupational discomfort, even fear of losing a job. These may turn into stressors that are embedded in the immediate work environment if there are not enough resources to cope with them, which in turn lead to unfavorable work consequences (Mäkiniemi *et al.*, 2021; Meyer *et al.*, 2021; Shao *et al.*, 2021).

The job demands-resources (JD-R) model (Bakker and Demerouti, 2017) and the conservation of resources (COR) theory (Hobfoll, 2001) are used as theoretical frameworks in the current study to explain WFH in pandemic-related situations. According to the JD-R model, remote workers use their personal resources to deal with job demands. They may experience emotional exhaustion if they are confronted with excessive demands that require their efforts or resources to the extent that they find it difficult to recover. COR builds the theoretical model for work-home resources, especially in the context of crises and demanding situations such as WFH during lockdowns (Hobfoll, 2010; Kumar *et al.*, 2021). The theory suggests that individuals will endeavor to acquire and maintain their reservoir of resources (Hobfoll, 2001). However, individuals do not have an infinite number of resources, which they need to expend to meet the demands of their organizational obligations and personal lives (Halbesleben *et al.*, 2014; ten Brummelhuis and Bakker, 2012). Stress is a reaction to an environment where people experience a threat or an absence of resources. Thus, the loss of or threat to lose these resources may lead to strain (e.g. emotional exhaustion, depression) among remote workers, which in turn leads to negative outcomes (e.g. dissatisfaction, turnover intention) (Halbesleben, 2006; van Woerkom *et al.*, 2016).

It is interesting to note that the extents and effects of WFH on employees' attitudes, perceptions, and work behaviors are still unpredictable and could pose a significant issue in the era of the new normal. This sudden shift to remote work may have changed the way remote workers perceive their work life and the resources they need to deal with the changing demands. Studies conducted before the COVID-19 pandemic may not appropriately reflect the current situation, raising the question of how this work approach and demanding situations

may affect employees' work and lives over time and which stressors are most pertinent (Ahlstrom and Wang, 2021; Chen, 2021; Kumar *et al.*, 2021). Moreover, prior research on WFH during the initial lockdowns has typically focused on a single stressor (e.g. job instability, social isolation) at a time, which may not adequately capture various stressors for comparison (e.g. Belzunegui-Eraso and Erro-Garcés, 2020; Zheng *et al.*, 2020). This study then expands on the investigation of multiple stressors possibly caused by WFH in the new normal.

Grounded in this backdrop, this study aims to highlight WFH stressors (i.e. home-work conflict, perceived insufficient organizational support, and perceived social isolation) identified from the lists of exhaustive resources during crises (Hobfoll, 2010) and prior studies conducted in the earlier lockdowns (e.g. Chen, 2021; Meyer *et al.*, 2021; Shao *et al.*, 2021; Syrek *et al.*, 2022; Zheng *et al.*, 2020) and investigate whether these stressors may intensify emotional exhaustion. It also examines whether they transmit their effects via emotional exhaustion as a mediator to work-related (i.e. job embeddedness) and non-work related (i.e. life satisfaction) outcomes over the course of the pandemic. Given that the effect of WFH might vary depending on the context of the country, this study then provides insights from Thailand, where online remote working was relatively new when the full-scale lockdowns were initially implemented and where knowledge on how employees are reacting to this change is fairly limited. Findings from this study should help managers and practitioners raise awareness of employees' perspectives, redesign human resource management (HRM) measures to prioritize reactions toward their work and life during difficult times, and build up resources to help individuals cope with the enduring and demanding work situations such as crises. Importantly, explaining possible risk factors at work could also help organizations create effective strategies for preventing these unprecedented and undesirable consequences, specifically in the aftermath of the crisis or when facing a critical event like this again.

Theoretical background and hypothesis development

The underlying theories, the pandemic, and WFH

This study builds upon the JD-R model (Bakker and Demerouti, 2017) and the COR theory (Hobfoll, 2001). The former suggests that every occupation has its own factors associated with job stress, which can be classified into two distinct categories, namely job demands and job resources. Job demands consist of physical, psychological, organizational, or social aspects, which require individual vitality. Putting in high levels of effort to the extent that employees find it too hard to recover could result in both physical and mental costs, which can lead to fatigue, exhaustion, or irritation. Job resources are characterized as physical, psychological, organizational, or social aspects that reduce stress and burnout due to excessive job demands, promote employee wellbeing and satisfaction, and may serve as safeguards against unfavorable outcomes.

The central premise of COR is that people have an innate drive to acquire resources and resist resource loss (Hobfoll, 2001). The concept employs an investment-focused perspective, arguing that employees not only seek to protect their existing resources, but also seek to make investments that help them gather resources they can use in the future. Resources can be defined as personal characteristics, energies, and conditions that are perceived as important themselves or that individuals centrally value within the organizational context. Positive work outcomes (e.g. satisfaction, wellbeing, retention) are likely to occur when individuals perceive that they have appropriate or sufficient resources for coping with a certain situation, while negative outcomes (e.g. stress, dissatisfaction, quitting) are the result of resource depletion or insufficiency over time without replenishment (Hobfoll, 2010). The theory also posits that resource losses are more salient than resource gains and that individuals must invest resources to gain resources, to protect themselves from losing resources, and for recovering from losing resources (Halbesleben *et al.*, 2014).

During the pandemic crisis, organizations have opted for a remote work approach to minimize employee exposure to the public and reduce the spread of the virus. Individuals are urged to stay at home and practice social distancing to limit face-to-face interactions with others. In response to such circumstances, the homes of workers suddenly became the main place of economic activity (Vyas and Butakhieo, 2021). Prior studies found that many workers are not capable of adapting to the new office environment through the virtual space of the Internet (e.g. Chen, 2021; Syrek *et al.*, 2022). The amount of available support, particularly from the employer, is either overlooked or undermanaged (Shao *et al.*, 2021). The pandemic-related demands and changing work situations, as well as the shortages of supportive resources, tend to have a major detrimental impact on workers' lives and psychological states of mind (McQuaid *et al.*, 2021; Syrek *et al.*, 2022). The sudden changes to the new work arrangements, without previous experiences and insufficient resources to deal with them, is likely to deplete individuals' emotional and physical energy reserves over time (Bakker and Demerouti, 2017; Hobfoll, 2010) and subsequently cause stress and exhaustion (Halbesleben, 2006; van Woerkom *et al.*, 2016). These may turn into work stressors (Meyer *et al.*, 2021; Shao *et al.*, 2021), which cumulatively drain individuals' resources, signifying future losses (Hobfoll, 2010). Such losses may then lead to reduced physical and mental health, dissatisfaction, and turnover intention (Halbesleben *et al.*, 2014). It can therefore be said that both of the aforementioned theories appear especially pertinent for predicting the unfavorable outcomes of WFH in a threatening and stressful event such as the pandemic because they address the interactions of demands and resources as well as the related temporal dynamics when employees encounter crises.

The pandemic and WFH in Thailand

Thailand's social and economic development has advanced significantly over the past few decades, and as of 2011, its economy has been classified as upper-middle income. The country is making progress towards achieving its sustainable development goals and aims to become a high-income nation by 2037 (World Bank, 2021). According to Hofstede *et al.* (2010), Thailand is a highly collectivist nation. This cultural dimension is characterized by having a rigid social structure, placing a high priority on relationships, and emphasizing the group as a whole, as can be seen in the emphasis on reciprocal support and connections in close groups. Social gatherings, living with and for others, and working together as a group could be anticipated in society. Thais tend to accommodate changing circumstances, particularly among their in-groups (i.e. family, work team), to support and maintain their close relationships and harmony. This specific norm may, therefore, have an implicit impact on how Thai organizations and workforces respond to the pandemic's abrupt changes and challenging work environments.

Thailand's COVID-19 infection rates during the outbreak in the middle of 2020 were comparatively low, the spread was slow, and the majority of cases registered were locally transmitted (Ministry of Public Health of Thailand, 2020). However, the Thai Government ramped up total lockdowns to contain the virus's spread following the directives of the World Health Organization. An emergency decree to control the pandemic was announced, enforcing people to self-quarantine, refrain from meetings, practice social distancing, and work from home. Due to the strict social measures, about 70% of Thai employees began working remotely for the first time (Bangkok Post, 2021). It might be stated that WFH was fairly new to Thailand during that time and that both employers and employees did not find the concept to be particularly appealing, as its economy is less digitalized than other developed countries. Thailand, like most countries in the emerging market, has long followed the standard practice of working in a formal office environment. They are required to work at the office during regular office hours, with no flexible working options being offered.

Only a small proportion of employees had the experience and opportunities to work remotely before the outbreak (Bangkok Post, 2020). But the pandemic crisis made this work arrangement necessary. While some people were able to adapt on their own to the sudden changes, many others were unprepared, and not everyone was willing to. The new work approach seemed to influence Thai employees' lives, personal and work obligations, relationships, and wellbeing, like in most countries. However, prior research suggests the impact of WFH could vary depending on the characteristics of each country (e.g. Afrianty *et al.*, 2021; Meyer *et al.*, 2021). What happens in Thailand, as an emerging economy with a highly collectivist culture, may differ from those found in Western economies or other countries. Perspectives from Thailand may additionally contribute to the body of relevant research and the significance of the study.

WFH stressors and emotional exhaustion

Emotional exhaustion is a state of feeling worn-out and drained as a result of accumulated stress from personal or work lives, or a combination of both (Maslach and Jackson, 1981). Based on COR, which specifies that stress occurs when resources are depleted (Halbesleben, 2006; Hobfoll, 2001), emotional exhaustion is most likely to happen when there is an actual or threatened resource losses, when resources are inadequate to meet work-related demands, or when resource returns are less than resource investment. During the pandemic, WFH brings flexibility and convenience for some, while for many, working remotely becomes stressful, exhausting, and overwhelming (Belzunegui-Eraso and Erro-Garcés, 2020).

To identify the core and possible antecedents of negative psychological states of mind (e.g. emotional exhaustion) for further analysis, the following selection approaches are considered for this study. First, the lists of 74 exhaustive personal resources in crises (Hobfoll, 2010) were used as a guiding principle, and the four main groups from the inventories: social support, objects or conditions, constructive resources, and energies (Halbesleben *et al.*, 2014) were taken into account. Second, the interview results from the research by Shao *et al.* (2021) pinpointing the stressors experienced by employees at work during the outbreak were evaluated. The findings identify four categories of pandemic-related work stressors, including work-family boundary stressors, technology stressors, work coordination stressors, and workload stressors, that leave WFH staff feeling worn out. Third, human resource (HR) practitioners' perspectives from Chen's (2021) survey were analyzed. These practitioners view conflicts with family, personal failure to adapt to a remote working approach (e.g. feeling isolated, social disconnection), and a lack of organizational support (e.g. teleworking skills, equipment) as major work stressors for WFH employees. Fourth, the current study found that ambiguous home-work boundaries, demands for constant availability, dependence on digital communications and a variety of technological platforms, a lack of organizational support, and a lack of social connections have been cited as central in several studies during the initial phases of the pandemic (e.g. Belzunegui-Eraso and Erro-Garcés, 2020; Mäkiniemi *et al.*, 2021; McQuaid *et al.*, 2021; Meyer *et al.*, 2021; Syrek *et al.*, 2022; Zheng *et al.*, 2020). These stressors have a propensity to drain individuals' energy and resources over time, and the depletion of resources is likely to worsen emotional weariness. Lastly, this study suggests that the selection criteria should make the case that HR practitioners can intervene and recover such pandemic-related work stressors, especially those that could benefit the processes of employee support, motivation, and development. Taking all criteria together, the current research therefore proposes that family-work conflict, perceived insufficient organizational support, and perceived social isolation can be important stressors induced by WFH and the crisis.

WFH employees are required to use family or home space for their own work activities during lockdowns. The abrupt changes in both work-related and private demands without

the necessary home-office skills create a conflict between work and family role boundaries (Chen, 2021). When individuals work at home, some family tasks, such as childcare or housework, need to be shared between them (Vyas and Butakhieo, 2021). Role theory (Kahn *et al.*, 1964) suggests that individuals hold and maintain a variety of roles on a daily basis. The various roles may be inherently incompatible due to certain obligations and expectations. This imbalance creates inter-role conflict, which may occur when the demands associated with participation in one role interfere with one's ability to participate in and manage the situations of other roles. If work roles interfere with an individual's ability to fulfill family or home roles, conflict between work and home exists, and such conflict can also flow from family or home to work because home role demands interfere with one's ability to fulfill work demands (Grandey and Cropanzano, 1999; Netemeyer *et al.*, 1996). According to the JD-R and COR theories, WFH employees are compelled to use and re-allocate their limited resources from one domain to meet the demands of the other, while replenishment of resources during the imbalance situations does not always occur (Bakker and Demerouti, 2017; Hobfoll, 2001; Meyer *et al.*, 2021). Not surprisingly, this work stressor is more likely to occur and interfere with work activities in the home setting, as it involves more frequent interactions between the employee and his or her family members. Such demanding situations with insufficient resources eventually lead to emotional exhaustion (Chen, 2021; Kumar *et al.*, 2021).

As suggested by Peltokorpi (2020), this study uses the term "*home*" instead of "*family*" as a means of studying the general incompatibility between employees' work and nonwork demands in order to better encompass the wider engagement of not only family members but also friends, partners, and local communities. The current research focuses on home-work conflict, which occurs when the general demands of time devoted to and strain created by home-related issues interfere with performing work-related duties, even though work-home conflict is more common and has gained more research attention (Williams *et al.*, 2016). This argument is valid for the impacts of daily online remote work due to WFH arrangements on home interference, particularly the intervention at home and the occupation of family members' space during lockdowns.

Perceived organizational support captures employees' beliefs regarding the extent to which the organization cares for their wellbeing and provides access to job-related resources as necessary (Eisenberger *et al.*, 2002). According to the underlying theories, organizational support is a resource that can protect employees against the harmful effects of both work and non-work demands (Bakker and Demerouti, 2017; Hobfoll, 2010). A high level of support from the employer may alleviate the impact of such demands by helping employees to cope with difficult barriers and be able to run a smooth workflow. During the pandemic, many organizations have implemented full-time remote work for their employees in response to this critical situation, and employees may expect that their employer should provide them with necessary and adequate resources, such as relevant work equipment and materials, information technology and communication mechanisms, timely information, and work procedures to support the WFH model (Belzunegui-Eraso and Erro-Garcés, 2020; Chen, 2021; Syrek *et al.*, 2022). Without appropriate support and sufficient resources, workers may find it difficult to handle unplanned situations and perform effectively in this online remote working scenario. They need to increasingly struggle to fulfill the obligations at work and home by themselves, with limited resources, and perhaps with their inability to manage existing resources (ten Brummelhuis and Bakker, 2012; Halbesleben *et al.*, 2014). This represents a potential threat of resource loss, and subsequently leads to stress and fatigue (Bakker and Demerouti, 2017). Recent studies have indicated that when employees perceive that their organization provides adequate resources to support WFH during lockdowns, they report higher psychological health and fewer negative outcomes (e.g. Kumar *et al.*, 2021; Meyer *et al.*, 2021). Conversely, it can be argued that insufficient job-related resources provided by the employer may amplify the burden of job demands, making the situation even worse

(Mäkiniemi *et al.*, 2021). That is, employees with lower levels of perceived organizational support are likely to demonstrate higher levels of emotional exhaustion.

The introduction of WFH and physical distancing requirements instituted to control the infection has significantly reduced the frequency of social contacts and of face-to-face interactions at work and leisure. Such a massive scale of social distancing may impact the basic human need for social belonging, particularly in the collectivist fabric of Thai society, and subsequently raise the likelihood of social isolation (McQuaid *et al.*, 2021; Mäkiniemi *et al.*, 2021). Perceived social isolation is defined as the distressing subjective feeling of lacking a social network or companions and refers to the perception that one's social relationships are inadequate (Holt-Lunstad *et al.*, 2015). This perception has often been put forward as the biggest disadvantage of remote working, with severe impacts in the case of full-time WFH, since it significantly curbs opportunities for social engagements among employees (Chen, 2021; Golden *et al.*, 2008). It is possible that this kind of stressor may spill over into the work domain and make it more difficult for some employees to adapt to the changing circumstances, increasing the risk for emotional exhaustion. This is because social interactions as job-related resources are not adequate to deal with such difficulties, and certain situations are so overwhelming that the resources available to teleworkers may not be enough. In both cases, an irreparable deterioration of resources would occur (Halbesleben, 2006; Holt-Lunstad *et al.*, 2015).

Although WFH employees can virtually interact with colleagues online during lockdowns, they may miss networking opportunities and be unable to fulfill the basic human need for social belonging. Importantly, communicating and working through electronic means on a daily basis limits the richness of information and prevents the sharing of interpretive contexts, social cues, or body languages, all of which are essential for the mutual understanding of both professional and personal needs (Shao *et al.*, 2021; Wiesenfeld *et al.*, 1999). Face-to-face interactions, on the other hand, might make it easier to find common ground and enable people to get feedback promptly, lessening feelings of isolation and mental stress. Prior empirical research conducted in the early stages of the pandemic also emphasizes how social isolation is typically linked to increased levels of stress, decreased psychological wellness, and decreased life satisfaction (e.g. Cacioppo and Cacioppo, 2014; Mäkiniemi *et al.*, 2021; Zheng *et al.*, 2020). It can be assumed that the lack of regular face-to-face interactions and social engagements while working online is likely to cause feelings of loneliness, which raises the possibility of emotional exhaustion.

In sum, the current study argues that employees feel emotionally exhausted, while trying to deal with demands associated with home-work conflict, struggling with inadequate support from their employer, and feeling isolated during lockdowns and the extension of WFH. Therefore, the following hypotheses are proposed:

H1. Home-work conflict is positively related to emotional exhaustion.

H2. Perceived insufficient organizational support is positively related to emotional exhaustion.

H3. Perceived social isolation is positively related to emotional exhaustion.

Emotional exhaustion, job embeddedness, and life satisfaction

Job embeddedness signifies the general factors that affect employees' decision to keep their post (Mitchell *et al.*, 2001), and it is particularly related to why employees want to continue their jobs (Karatepe, 2013). The stronger the connections individuals hold with others in the workplace, the more they are bound to their job and organization, thus inducing them to stay, despite opportunities presenting themselves elsewhere (Crossley *et al.*, 2007). Based on the assumptions of COR (Hobfoll, 2001), affective bonding is one of the motivating factors

of attachment and withdrawal, and workers use such affective reactions to communicate their psychological comfort or discomfort at work. Individuals with comfort are likely to have the intention to stay, while those with discomfort are likely to leave (Holtom *et al.*, 2012).

In the context of the pandemic, this study posits that WFH employees may not be tightly connected to the organization, because they cannot cope with changes and difficulties surfacing from work stressors (i.e. home-work conflict, loneliness, insufficient organizational support) and emotional exhaustion due to the lack of links with other people in the workplace. They may not feel attached to the organization, since they find that their personal values, needs, career goals, and skills do not fit well with the demands of the jobs and the changing work approach. Importantly, working in difficult situations without adequate planning and resources to handle with them is likely to drain employees' emotional and physical energy reserves (Halbesleben *et al.*, 2014; Hobfoll, 2010). Encountering a threat to resources, they appear to exhibit high levels of emotional exhaustion and low levels of job embeddedness, as reported in prior studies (e.g. Karatepe, 2013; Peltokorpi, 2020).

Life satisfaction refers to the feeling of wellbeing with oneself in one's surroundings (Erdogan *et al.*, 2012). It reflects a cognitive judgment about life quality according to self-set standards and criteria. In accordance with COR (Hobfoll, 2001) and work-life balance research (Crosbie and Moore, 2004), this study argues that emotional exhaustion should have a negative relationship with life satisfaction. For most employees, work and home represent the two primary domains of life. Work occupies a significant portion of their waking day. When work is out of balance due to difficult times or distress, there should be a spillover effect on an individual's life overall. If employees are stressed and feel unhappy with online remote work during the nationwide lockdowns, the overall quality of their lives in general should be affected (Chen, 2021; Kumar *et al.*, 2021). Feelings of being emotionally overextended and exhausted due to the high demands and significant consumption of personal resources may lead to discord in workers' lives, which then would reduce their overall life satisfaction. This is because individuals have few resources for other activities in their non-work related domain, thus diminishing their satisfaction with life. Previous research also states that emotionally exhausted workers would be less likely to report high levels of life satisfaction (e.g. Boekhorst *et al.*, 2017; Meyer *et al.*, 2021). Based on all the above logic, it is argued that emotional exhaustion has an inverse relationship with life satisfaction.

This study investigates both job-related and life-related outcomes of emotional exhaustion by proposing that employees with elevated levels of emotional exhaustion are unlikely to be embedded in their job and satisfied with their life. The preceding discussion therefore leads to the following hypotheses:

H4. Emotional exhaustion is negatively related to job embeddedness.

H5. Emotional exhaustion is negatively related life satisfaction.

The mediating effects of emotional exhaustion

This study additionally expands the JD-R model (Bakker and Demerouti, 2017) and the COR theory (Hobfoll, 2001) for developing the relationships among emotional exhaustion as a mediator of the effects of home-work conflict, perceived insufficient organizational support, and perceived social isolation, also known as WFH stressors on work (i.e. job embeddedness) and nonwork (i.e. life satisfaction) outcomes. Prior studies posit that the presence of job demands and the lack of support consume employees' physical and emotional resources and therefore intensify emotional exhaustion, which in turn leads to unfavorable employee outcomes (e.g. Boekhorst *et al.*, 2017; Halbesleben *et al.*, 2014; Karatepe, 2013; Kumar *et al.*, 2021). Employees who experience conflict between home and work roles, perceive that their employer's support is insufficient, and feel isolated while working remotely from home

for an extended period of time during the pandemic are likely to experience high levels of stress. These stressors make individuals more susceptible to suffering other resource losses during the crisis, and subsequently becoming emotionally exhausted. If workers cannot cope with emotional exhaustion, the consequential depressive mental state may pave the way to develop negative thoughts about their job, leading to the idea of leaving as a result (Halbesleben *et al.*, 2014; Holtom *et al.*, 2012; Karatepe, 2013), and the effect will also lower their life satisfaction (Kumar *et al.*, 2021; Meyer *et al.*, 2021). During the long lockdown period, it can be assumed that WFH stressors can cumulatively transmit their effects through emotional exhaustion as a mediator and create unfavorable outcomes. In this case, emotional exhaustion may shed important light on the underlying mechanisms by which WFH stressors lead to negative consequences. Understanding the mediating role of emotional exhaustion can assist individuals and organizations in determining the root causes of such unfavorable outcomes and develop targeted interventions to address them appropriately. Thus, the current study goes one step further in the field by investigating how WFH stressors indirectly impact both work and non-work outcomes via emotional exhaustion. Within the context of the relations between the variables and the abovementioned explanations, the following hypotheses are proposed:

H6. Emotional exhaustion mediates the effects of (a) home-work conflict, (b) perceived insufficient organizational support, and (c) perceived social isolation on job embeddedness.

H7. Emotional exhaustion mediates the effects of (a) home-work conflict, (b) perceived insufficient organizational support, and (c) perceived social isolation on life satisfaction.

Methodology

Sample and procedures for data collection

This study employed a time-lagged survey study administered electronically through anonymous self-report questionnaires. Data was collected in two waves (Time 1 and Time 2), and the time lag between each wave was four weeks. These time lags were long enough to reduce concerns about recall bias, but not too long that significant organizational events could have occurred during the data collection. Collecting data in different waves also diminishes the likelihood of common method variance (CMV) due to self-report data (Podsakoff *et al.*, 2012).

This survey was administered to full-time adult employees who experienced the WFH model during the nationwide lockdowns. A convenience and nonprobability sampling method was utilized through the snowball technique (Goodman, 1961). One hundred MBA students from two major business schools in Bangkok, Thailand, were invited to take part in the study as recruiters, and 82 voluntarily participated. Each participating student was asked to recruit up to 15 full-time employees (i.e. friends, colleagues, or acquaintances) who had been working from home at that time to participate in this survey. The recruits provided their names and e-mail addresses to the recruiting students, who passed them on to the researcher. The names and e-mail addresses of these individuals were compiled, and a survey invitation, including the research objectives and procedures, and the Time 1 survey links were then e-mailed. One week after the initial invitation, a reminder message was sent to these invited participants. The Time 1 questionnaire contained WFH stressors (i.e. home-work conflict, perceived social isolation, and perceived insufficient organizational support) and demographic profiles (i.e. gender, age, tenure, marital status). At the first time point, a total of 651 questionnaires had been completed, representing a response rate of 53%.

At the second time point, the Time 2 questionnaire links were only sent to those who had participated at Time 1. The second wave survey captured outcome variables (i.e. emotional exhaustion, job embeddedness, and life satisfaction). Participants' e-mail addresses were used

as identification codes to match the surveys from both waves. In each survey round, participants were asked to complete the questionnaires as part of an academic research project. They were informed that their complete confidentiality was guaranteed, that involvement was voluntary, and that they could withdraw from the study at any time without any negative consequences. This study was approved by the Research Ethics Review Committee for Research Involving Human Subjects (Certificate No. 215/2564). Of those solicited to participate at the second time point, 422 employees completed this survey, resulting in a response rate of 64%, of which 418 were useable, representing a cumulative response rate of 34%.

In the final sample used for hypothesis testing, the mean age of the respondents was 34.84 (SD = 8.09) and mean tenure was 6.04 (SD = 6.00). Three-fourths (74.6%) of the participants were female. A total of 72% were single, 27% were married, and the rest were separated.

Measures

Measures for the studied constructs were developed from prior literature. All the measuring instruments used five-point Likert scales, ranging from "strongly disagree (1)" to "strongly agree (5)." The questionnaire was initially developed in English, translated into the local language, and then back-translated into English to ensure equivalency of meaning (Brislin, 1980).

Home-work conflict (Time 1). This construct was measured by a five-item scale from (Netemeyer *et al.*, 1996). An example statement is, "The demands of my family members interfere with work-related activities." Higher scores indicate higher levels of conflict between family or home and work felt by workers.

Perceived insufficient organizational support (Time 1). This study measured perceived organizational support using a six-item scale taken from Rhoades and Eisenberger (2002). As suggested by (Mäkiniemi *et al.*, 2021), some items were rearranged to reflect the pandemic and WFH events. A sample item is, "I have received enough instructions on performing my tasks and duties from home." To facilitate the interpretation of the results, each item was reverse-scored, so higher scores represented lower levels of organizational support.

Perceived social isolation (Time 1). To capture a sense of isolation at work experienced by employees, this construct was assessed with seven items from Golden *et al.* (2008). A sample item is, "I feel left out of activities and meetings that could enhance my career." A higher rating indicates a higher level of perceived isolation.

Emotional exhaustion (Time 2). This construct was measured using the Maslach Burnout Inventory (Maslach and Jackson, 1981). The eight-item scale captures feelings of being emotionally overextended and exhausted by one's work. An example statement is, "Working from home all day is really a strain for me." Higher scores correspond to greater experienced burnout.

Job embeddedness (Time 2). The scale to measure embeddedness was adapted from the global "*on-the-job*" embeddedness items developed by Crossley *et al.* (2007). A sample item is, "I am tightly connected to this organization." A higher rating indicates a higher level of embeddedness.

Life satisfaction (Time 2). The items were taken from Satisfaction with Life Scale (SWLS) developed by Diener *et al.* (1985). A sample item is, "The conditions of my life are excellent." A higher rating indicates a higher level of life satisfaction.

Demographic control variables (Time 1). Gender was measured using a dichotomous variable, where 0 indicates a male and 1 indicates a female respondent. Marital status was measured through the use of self-reported items via a three-point scale, denominated as single, married, and divorced or separated. Participants' age and tenure were captured in years.

Measurement model and assessment of common method variance

Before hypothesis testing, a confirmatory factor analysis (CFA) was conducted to validate all multiple-item scales to indicate how well the identified measures predict the latent variables

(Byrne, 2010). The CFA results indicate that the six-factor model (i.e. home-work conflict, perceived social isolation, perceived insufficient organizational support, emotional exhaustion, job embeddedness, and life satisfaction) shows acceptable fit to the data (see Table 1): χ^2/df = 2.13, goodness-of-fit index (GFI) = 0.88, normed fit index (NFI) = 0.90, Tucker–Lewis index (TLI) = 0.94, comparative fit index (CFI) = 0.94, and root means square error of approximation (RMSEA) = 0.05 (Hu and Bentler, 1999). These indicators all significantly loaded on their respective latent factors.

When all items were combined and loaded into a CFA to test whether a single factor might account for the majority of the covariance among the measures, the one-factor model showed a poor data fit: χ^2/df = 13.65, GFI = 0.26, NFI = 0.31, TLI = 0.28, CFI = 0.32, and RMSEA = 0.17. Other measurement models were tested by loading the related observed variables into a three-factor model, a four-factor model, and a five-factor model respectively. All comparative models and the chi-square difference tests demonstrate worse fit than the six-factor model, suggesting that the majority of the variance in this model was not explained by a single factor. In addition, the analysis of a common latent factor (CLF) was conducted, and its related standardized regression weights of all items for models with and without CLF were compared. The differences in these regression weights were found to be very small ($\Delta < 0.2$) (Gaski, 2017). These analyses and findings indicate that the constructs are distinct from one another, and that CMV does not seem to be a pervasive problem in this data set (Podsakoff *et al.*, 2012).

Convergent validity was tested using factor loadings, composite reliability (CR), and average variance extracted (AVE) (Hair *et al.*, 2010). As shown in Table 2, convergent validity was supported, because standardized coefficients from items to factors ranged from 0.64 to 0.87, with statistically significant regression weight, and all items exceeded the conventional threshold of

Measurement models	χ^2/df	GFI	NFI	TLI	CFI	RMSEA
1. One-factor model	13.65	0.26	0.31	0.28	0.32	0.17
2. Three-factor model (Home-work conflict + Perceived insufficient OS + Perceived social isolation, Emotional exhaustion, Job embeddedness + Life satisfaction)	9.27	0.43	0.53	0.53	0.56	0.14
3. Four-factor model (Home-work conflict, Perceived insufficient OS, Perceived social isolation, Emotional exhaustion + Job embeddedness + Life satisfaction)	8.15	0.47	0.59	0.60	0.62	0.13
4. Five-factor model (Home-work conflict, Perceived insufficient OS, Perceived social isolation, Emotional exhaustion, Job embeddedness + Life satisfaction)	4.76	0.69	0.76	0.79	0.80	0.09
5. Six-factor model (All variables)	2.13	0.88	0.90	0.94	0.94	0.05

Note(s): OS = Organizational support
Source(s): Author's work

Table 1.
Comparison of measurement model fit indices

Constructs	Factor loading range	α	CR	AVE
Home-work conflict (5 items)	0.77–0.82	0.89	0.89	0.63
Perceived social isolation (7 items)	0.64–0.76	0.88	0.88	0.51
Perceived insufficient OS (8 items)	0.72–0.87	0.94	0.94	0.66
Emotional exhaustion (8 items)	0.69–0.87	0.94	0.92	0.60
Job embeddedness (7 items)	0.72–0.84	0.93	0.91	0.60
Life satisfaction (5 items)	0.68–0.78	0.89	0.85	0.53

Note(s): OS = Organizational support
Source(s): Author's work

Table 2.
CFA and validity of the measurement model

0.50 (Anderson and Gerbing, 1988). CR values, which illustrate the degree to which the construct indicators indicate the latent variables, ranged from 0.85 to 0.94. This exceeds the recommended value of 0.70 (Hair *et al.*, 2010). The AVE values, which reflect the overall amount of variance in the indicators representing the latent construct, also exceeded the recommended value of 0.50 (ranging from 0.51 to 0.66), indicating adequate convergent validity (Bagozzi and Yi, 1988). As shown in Table 3, all the square roots of the AVE values displayed on the diagonal were greater than the correlation values in the same row and column, legitimizing adequate discriminant validity (Fornell and Larcker, 1981). Cronbach's alphas ranged from 0.88 to 0.94 and were adequate for all constructs.

Results

Descriptive statistics

Descriptive statistics and the bivariate correlations for the related constructs are presented in Table 3. These intercorrelations provide preliminary support for the hypothesized relationships. An initial investigation of the results revealed some high correlations ($r > 0.50$), which indicate the potential issue of multicollinearity. The variation inflation factors (VIF) were further examined to check for multicollinearity of the focal variables. No issue was detected since all VIF values were lower than 5, below the conventional cutoff of 10 (Hair *et al.*, 2010). Thus, all variables were retained for regression analyses.

Hypothesis testing

Structural equation modeling (SEM), with the maximum likelihood method of estimation, was performed to test the proposed hypotheses and multiple mediation models using AMOS software version 28.0. All analyses were conducted with gender, age, tenure, and marital status as the control variables. The path coefficients associated with the hypothesized model are reported in Figure 1 and Table 4.

As shown, home-work conflict, perceived insufficient organizational support, and perceived social isolation were all significantly and positively related to emotional exhaustion ($\beta = 0.32$, 0.25, and 0.49, $p < 0.01$ respectively), providing support for H1, H2, and H3. The variables in the model explained 51% of the variance in emotional exhaustion. Along with the findings, emotional exhaustion was found to significantly and negatively influence job embeddedness ($\beta = -0.17$, $p < 0.01$) and life satisfaction ($\beta = -0.28$, $p < 0.01$), explaining 31% and 39% of the variance in both outcome variables, respectively. Hence, there is empirical support for H4 and H5.

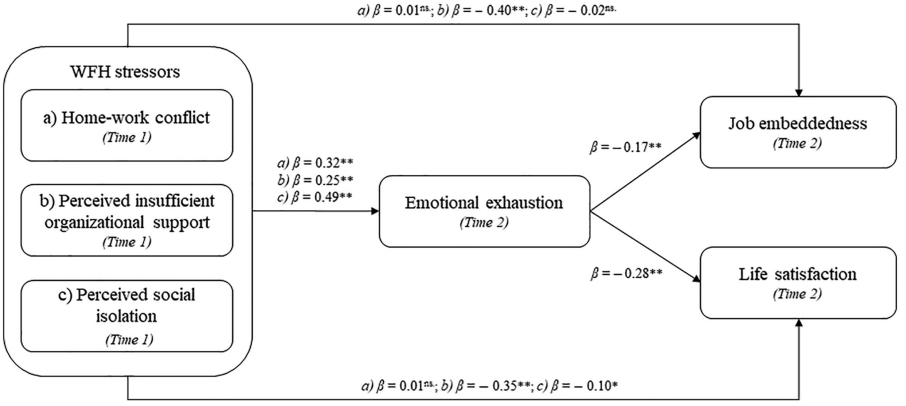
To test the mediation hypotheses, the total effects of the three WFH stressors (i.e. home-work conflict, perceived insufficient organizational support, and perceived social isolation) on the outcome variables (i.e. job embeddedness and life satisfaction) were initially examined without the presence of the mediator (i.e. emotional exhaustion). The estimates of the paths from perceived insufficient organizational support to job embeddedness and life satisfaction were negatively significant ($\beta = -0.44$ and -0.43 respectively, $p < 0.01$). Also, negative and significant relationships between perceived social isolation and both outcome variables were found ($\beta = -0.10$, $p < 0.05$ and -0.17, $p < 0.01$ respectively). However, home-work conflict failed to reach significance ($\beta = 0.03$ and 0.02 respectively, $p > 0.1$). Regardless of whether the total effect is statistically insignificant, scholars suggest that it is still legitimate to continue examining indirect effects (Hayes, 2018; Rucker *et al.*, 2011). Significant indirect effects could occur in the absence of significant total or direct effects (Zhao *et al.*, 2010).

User-defined estimands in AMOS were run to further test specific indirect effects in the hypothesized mediating paths. The bootstrapping procedure with 95% bias-corrected confidence estimates was also performed using 10,000 resamples to confirm such mediating effects and indirect relationships (MacKinnon *et al.*, 2004). The results indicate that

Table 3.
Descriptive statistics, intercorrelations, and square root of average variance extracted

Variables	1	2	3	4	5	6	7	8	9	10
1. Gender (Male = 0)	–									
2. Age (Years)	0.02	–								
3. Tenure (Years)	0.03	0.58**	–							
4. Marital status (Single = 0)	−0.01	−0.36**	−0.36**	–						
5. Home-work conflict	−0.07	−0.18**	−0.07	−0.07	*0.79*					
6. Perceived social isolation	−0.04	−0.23**	−0.16**	−0.14**	0.41**	*0.71*				
7. Perceived insufficient OS	0.04	−0.09	−0.02	−0.14**	0.02	−0.02	*0.81*			
8. Emotional exhaustion	0.07	−0.29**	−0.21**	−0.29**	0.44**	0.60**	0.21**	*0.78*		
9. Job embeddedness	−0.04	0.34**	0.30**	0.29**	0.05	−0.10*	−0.50**	−0.22**	*0.78*	
10. Life satisfaction	−0.08	0.26**	0.20**	0.24**	−0.07	−0.16**	−0.45**	−0.41**	0.61**	*0.73*
Means	0.74	34.84	6.04	0.29	2.65	3.37	2.41	3.22	3.16	3.10
Standard deviations	0.44	8.09	6.00	0.47	1.08	1.01	0.97	1.16	0.85	0.95

Note(s): *$p < 0.05$; **$p < 0.01$; Diagonals represent the square root of the average variance extracted (AVE); OS = Organizational support
Source(s): Author's work



Note(s): For reasons of brevity, the effects of control variables (i.e., gender, age, tenure, marital status) were not presented. *$p < 0.05$; **$p < 0.01$; ns = non-significant
Source(s): Author's work

Figure 1.
Path coefficients (direct effects)

Paths	Estimate	SE	95% CI lower	95% CI upper
H1. Home-work conflict → Emotional exhaustion	0.32**	0.05	0.23	0.43
H2. Perceived insufficient OS → Emotional exhaustion	0.25**	0.05	0.15	0.35
H3. Perceived social isolation → Emotional exhaustion	0.49**	0.06	0.38	0.61
H4. Emotional exhaustion → Job embeddedness	−0.17**	0.07	−0.29	−0.03
H5. Emotional exhaustion → Life satisfaction	−0.38**	0.07	−0.52	−0.24
H6a. Home-work conflict → Emotional exhaustion → Job embeddedness	−0.05*	0.02	−0.10	−0.01
H6b. Perceived insufficient OS → Emotional exhaustion → Job embeddedness	−0.04*	0.02	−0.08	−0.01
H6c. Perceived social isolation → Emotional exhaustion → Job embeddedness	−0.08**	0.04	−0.16	−0.02
H7a. Home-work conflict → Emotional exhaustion → Life satisfaction	−0.12**	0.03	−0.19	−0.07
H7b. Perceived insufficient OS → Emotional exhaustion → Life satisfaction	−0.09**	0.02	−0.15	−0.06
H7c. Perceived social isolation → Emotional exhaustion→ Life satisfaction	−0.19**	0.04	−0.19	−0.12

Note(s):*$p < 0.05$, **$p < 0.01$; SE = Standard error; CI = Confidence intervals; OS = Organizational support
Source(s): Author's work

Table 4.
SEM path analysis and bootstrapping

significant indirect and negative effects of the three WFH stressors on job embeddedness through emotional exhaustion were observed (home-work conflict: $\beta = -0.05$, $SE = 0.02$, $CI = -0.10$ to -0.01, $p < 0.05$; perceived insufficient organizational support: $\beta = -0.04$, $SE = 0.02$, $CI = -0.08$ to -0.01, $p < 0.05$; and perceived social isolation: $\beta = -0.08$, $SE = 0.04$, $CI = -0.16$ to -0.02, $p < 0.01$). As these confidence intervals did not include zero, it is confirmed that there are mediation effects to report. Therefore, H6a, H6b, and H6c are supported. The direct effect of perceived insufficient organizational support on job embeddedness was weakened but still significant after controlling the mediator ($\beta = -0.40$, $p < 0.01$), indicating partial mediation. Whereas the direct pathways from

home-work conflict and perceived social isolation to job embeddedness became non-significant ($\beta = 0.01$ and -0.02 respectively, $p > 0.1$), implying complete mediation.

The structural path and bootstrapping analysis also revealed that the relationships between the three WFH stressors and life satisfaction were significantly and negatively mediated by emotional exhaustion, as evidenced by a 95% bias-corrected bootstrap CI that did not include zero (home-work conflict: $\beta = -0.12$, $SE = 0.03$, $CI = -0.19$ to -0.07, $p < 0.01$; perceived insufficient organizational support: $\beta = -0.09$, $SE = 0.02$, $CI = -0.15$ to -0.06, $p < 0.01$; and perceived social isolation: $\beta = -0.19$, $SE = 0.04$, $CI = -0.19$ to -0.12, $p < 0.01$). All H7a, H7b, and H7c are thereby confirmed. The direct effect of home-work conflict and life satisfaction was insignificant after controlling the mediator ($\beta = 0.01$, $p > 0.1$), assuming full mediation. The direct pathways from perceived insufficient organizational support and perceived social isolation to life satisfaction were weakened but still significant ($\beta = -0.35$, $p < 0.01$ and -0.10, $p < 0.05$, respectively), indicating partial mediation.

Discussion

Using time-lagged data from Thai employees who experienced WFH during the pandemic lockdowns, the results of this study demonstrate that the three WFH stressors, namely home-work conflict, perceived insufficient organizational support, and perceived social isolation, significantly cause emotional exhaustion, and that these stressors are negatively and indirectly associated with job embeddedness and life satisfaction via the mediation of emotional exhaustion. That is, employees who feel exhausted in their work during lockdowns are less apt to be satisfied with their life and may feel less inclined to stay with their workplace.

Perceived social isolation has the largest direct effect on emotional exhaustion. It also appears that the indirect effect of perceived social isolation is the strongest among all stressors, and that the negative influence is stronger on non-work (i.e. life satisfaction) than work (i.e. job embeddedness) outcomes. This is because the unprepared changes to the WFH approach limit usual face-to-face interactions, which may uncharacteristically diminish opportunities for socializing and networking. It may be challenging for workers in collectivist societies, like Thailand, who value working as a group, prefer interdependence, and frequently need support from others to work remotely alone. Some employees, who get used to physical office spaces and find it difficult to adjust to the new normal, may perceive these as hindrances that will cause negative reactivity and stress, subsequently intensifying undesirable outcomes.

Interestingly, the total effects of home-work conflict on both outcome variables were not significant, but the indirect effects via the mediator were. The results imply that employees may not leave their jobs or feel unhappy with life in general due to the incompatibility between their work demands and personal obligations. One explanation might be that most survey participants are single. In the short term, they may not perceive home-work conflict as a major interference. However, if such conflict requires high levels of their efforts to the extent that they find it difficult to maintain and recover, they may cumulatively experience strain and weariness over time. When employees are emotionally exhausted for a prolonged period of time, they tend to have lower life satisfaction and may consider quitting. It can be said that the effects of home-work conflict on both outcome variables are entirely transmitted through the mediator.

The findings observed in the present study are in line with the literature related to the pandemic-related work stressors (e.g. Kumar *et al.*, 2021; Meyer *et al.*, 2021; McQuaid *et al.*, 2021; Shao *et al.*, 2021) and also extend existing knowledge concerning the relationship between stress experienced at work, particularly in the context of critical and overextended events (Halbesleben, 2006; Hobfoll, 2010) such as the pandemic and the new normal. Applying the theories of JD-R (Bakker and Demerouti, 2017) and COR (Hobfoll, 2001), the results support the notion that the continued drain of resources in highly stressful situations can result in resource deterioration among workers. Perhaps the WFH stressors are so

overwhelming that, although the resources are adequate, individuals' levels of capabilities are still insufficient to cope with them, especially when facing a new and uncertain situation. In other words, they may not be able to allocate adequate or limited resources to deal with these stressors, thus increasing their levels of emotional exhaustion, which will eventually lead to unfavorable work and non-work outcomes.

Theoretical implications

The current study makes important theoretical contributions in several ways. First, it reinforces the applications of the JD-R and COR theories in the WFH scenario. The findings contribute to our understanding of how job demands and resources interact with employee strain (e.g. emotional exhaustion) during times of crisis, extending the traditional JD-R model (Bakker and Demerouti, 2017) beyond the normal work context. The results further support COR's temporal focus, which predicts that fewer resources indicate later losses (Hobfoll, 2010). This is particularly useful in the context of the pandemic crisis. Also, the findings add to the body of prior research by incorporating both work-related (job embeddedness) and non-work related (life satisfaction) outcomes into the COR framework. This contribution is consistent with previous literature and responds to a call for research on the expansion of COR-based mechanisms into areas of employees' life domain (e.g. Kumar *et al.*, 2021).

Second, this study synthesizes multiple WFH stressors due to the pandemic-related demands and draws comparisons between them. The findings help identify which stressors are more influential, impacting employees' work and lives over time. Third, the current research evaluates home-work conflict, although work-home conflict is more common (Williams *et al.*, 2016). It validates the argument concerning the effects of home-work imbalance under unusual circumstances such as lockdowns, which involve unprecedented interference in home life and occupation of family members' time and space by work demands.

Lastly, the context of Thailand, a collectivist country, along with the attitudes and perceptions of the relevant workers when facing the crisis, are added to the literature regarding online remote working, stress, wellbeing, and retention. It is possible that WFH issues will still remain unpredictable and continue to be a significant challenge long after the pandemic is over, even after employees are asked to return to their workplaces. If so, new interventions will be needed to maintain work productivity and foster healthy relationships at a distance. Organizations and academics may find the findings useful as a guideline to proceed with relevant research in diverse cross-cultural contexts to better understand the new normal way of working and its impacts.

Practical implications

This study's findings offer important managerial implications that translate into practice in five ways. First, WFH as a new normal way of working may continue even after the pandemic. Learning from past experiences and realizing how different kinds of critical situations shape the workplace will help managers and practitioners understand important remote work-related issues and redesign more effective HRM measures to mitigate the negative effects and better cope with the changes. Understanding such circumstances within the particular cultural context, such as collectivism in Thailand, can also shed light on potential difficulties that organizations may encounter and can direct efforts to localize and facilitate a smooth transition into the age of the new normal.

Second, in order to mitigate and prevent social isolation, especially in collectivist societies, organizations should provide their managers with the related resources and training necessary to adequately supervise and closely support their remote team members. The

relevant actions involve setting up virtual spaces for more interactions amongst workgroup members to assure adequate levels of professional involvement while continually fostering and enhancing interpersonal ties and maintaining interdependence. For example, this might be done by scheduling regular VDO chats with team members, such as morning meetings or afternoon coffee breaks, simply to check in and ask how everyone is doing for maintaining face-to-face routines that are adaptable to a remote realm. Employment assistance staff should be trained to recognize mental health issues and be able to offer online advice and therapy during difficult times. Such home-friendly support should replenish employees with more necessary resources to perform their tasks and minimize the threat of resource losses. These could assist in making the WFH experience a little less isolating.

Third, organizational support regarding technological equipment and infrastructure, access to necessary materials, and high ergonomic standards should be put in place, further developed, and properly refined to support remote working populations. Also, they need to receive additional training in teleworking and online office procedures along with virtual work communication skills to thrive in their current environment.

Fourth, learning how to manage remote work can decrease the perception of home-work conflict. Therefore, organizations should equip their employees with time management skills, enabling them to divide the two spheres of home and work. HRM policies should allow WFH employees to flexibly schedule their working hours to meet their personal needs and preferences. Also, a procedure of not working long hours when redesigning work schedules should be considered to promote work-life balance and effective time management. These could alleviate the excessive work hours caused by remote workers who work too many hours at home and minimize possible home-work conflicts.

Finally, although organizations should redesign their WFH model to minimize or eliminate mental health impacts on their workforce, the onus of maintaining the wellbeing also lies on the individuals themselves. So, it is recommended that whenever employees feel over-exhausted from their work, they should consider taking a proper break to re-energize themselves. If a personal initiative does not work well, professional counseling should be provided to help them relieve such a negative feeling. However, sometimes the feeling of exhaustion may occur due to the workaholic nature of individuals (Bakker *et al.*, 2013). In light of this, training interventions should be provided for these workers to develop self-observation competence and proactively manage themselves to reduce the negative effects of emotional exhaustion.

Limitations and directions for future research

This study has some limitations that suggest some directions for future research. First, in this study, data were obtained from WFH employees with a three-week time lag between two waves of survey to reduce the possibility of CMV. Although time-lagged studies present some evidence for temporal causality, it may not be sufficient to make causal inferences conclusively (Grandey and Cropanzano, 1999). It is possible that a reverse causal relationship may exist between the mediator and the outcome variables. With this realization, future research could adopt a more rigorous longitudinal design over a longer period of time than in this study. Also, the study variables could be measured separately at multiple time points (e.g. on a daily or weekly basis) to figure out the direction of causality and developmental trajectories.

Second, the data collection in this research was based on self-evaluation, which could give rise to CMV and artificially increase the correlations between variables (Podsakoff *et al.*, 2012), though self-report measures are appropriate in specific circumstances (Conway and Lance, 2010). It is recommended that future studies might collect data from multiple data sources (e.g. supervisors, peers) to reduce parameter estimation error.

Third, the current study investigated two important outcomes of WFH stressors via emotional exhaustion. Future research could further explore other potential significant work and non-work consequences, such as off-the-job embeddedness, organizational citizenship behavior, and perhaps physical and mental health impacts. These could provide managers and practitioners with critical information to better understand working situations under a particular crisis in order to prepare more supportive resources, redesign more appropriate interventions, and develop more healthy motivation.

Fourth, this research only analyzes the effects of WFH stressors on employee outcomes during the pandemic. Future studies might replicate the current research model with a larger sample size to further analyze and compare the results when WFH becomes the new norm for some organizations and jobs.

Finally, although this study is based on a comprehensive data set from an Asian country, caution should be used when generalizing the findings to other cultural contexts. In this regard, replication studies in other regions or countries should be considered to cross-validate the results of this study to be more culturally diverse.

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